

## **CERTIFICADO DE CALIBRACIÓN**

**Cliente:** Sociedad Minera Cerro Verde S.A.A.

**RUC:** 20170072465

**Dirección:** Cal. Jacinto Ibáñez Nro. 315 Urb. Parque Industrial, Arequipa.

**Equipo:** Monitor Estacionario

**Código SMCV:** 35-AIT-2106

**Área:** PLANTA MOLY C1

**Modelo:** TOXGARD II

**Tipo de sensor:** H2S

**N/S:** C24

**Fabricado por:** Mine Safety Appliances Co.

**Estado Inicial:** Operativo

**Estado Final:** Operativo

**Fecha de calibración:** 13/08/2025

**Próxima calibración:** 13/02/2026



**Realizado: Marjorie Costa Zambrano**  
Supervisor Operativo



**Revisado: Ing. Guillermo Cervetto.**  
Gerente General



**I. Lugar de Calibración:**

Sociedad Minera Cerro Verde.

**II. Condiciones Ambientales:**

|                  |            |
|------------------|------------|
| Temperatura      | 22°C       |
| Presión          | 1027.1 hPa |
| Humedad Relativa | 10%        |

**III. Gases y Rango de Medición:**

0-100 ppm

**IV. Patrones utilizados durante la calibración:**

| N°         | Patrón Utilizado | N/P    | Lote N°   | N° Certificado | Expiración |
|------------|------------------|--------|-----------|----------------|------------|
| Cilindro 1 | Oxígeno (O2)     | 479857 | 402469728 | 1138387390     | Abr-2030   |
| Cilindro 2 | H2S              | 467897 | 403357708 | 1140195588     | Jun-2027   |

**V. Resultados de Medición**

| Gas | Lectura Patrón | Lectura del equipo | % Error |
|-----|----------------|--------------------|---------|
| H2S | 40 ppm         | 40 ppm             | 0%      |



VI. Anexos

 **CALGAZ™**  
CALGAZ,  
A DIVISION OF AIRGAS USA LLC  
821 Chesapeake Drive,  
Cambridge, MD 21613  
USA Tel. 1-800-638-1197  
www.calgaz.com

**CERTIFICATE OF ANALYSIS**

Date: 04/03/2025  
Order Number: 1138387390  
Lot Number: 402469728

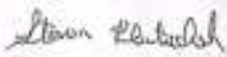
Customer: MSA DEL PERU SAC  
Part Number: 479857  
Use Before: APR 2030

| Component | Concentration<br>Accuracy: 02 ±5% |
|-----------|-----------------------------------|
| OXYGEN    | 20.8%                             |
| NITROGEN  | BALANCE                           |

Cylinder Size: 3.6 Cu. Ft.  
Contents: 103 Liter

Valve: 5/8"-18UNF  
Pressure: 1000 PSIG

Product composition verified by direct comparison to calibration standards traceable to N.I.S.T. weights and/or N.I.S.T. Gas Mixture reference materials.

Analyst:  
  
Steve Plutschak

  
Scott Tingle

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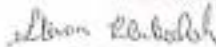
|                                 |                                   |
|---------------------------------|-----------------------------------|
| <b>Date:</b> 06/12/2025         | <b>Customer:</b> MSA DEL PERU SAC |
| <b>Order Number:</b> 1140195588 | <b>Part Number:</b> 467897        |
| <b>Lot Number:</b> 403357708    | <b>Use Before:</b> JUN 2027       |

| <b>Component</b> | <b>Concentration</b><br><b>Accuracy: H2S ±5%</b> |
|------------------|--------------------------------------------------|
| HYDROGEN SULFIDE | 40PPM                                            |
| NITROGEN         | BALANCE                                          |

**Cylinder Size:** 2.0 Cu. Ft.  
**Contents:** 58 Liter

**Valve:** 5/8"-18UNF  
**Pressure:** 500 PSIG

Product composition verified by direct comparison to calibration standards traceable to N.I.S.T. weights and/or N.I.S.T. Gas Mixture reference materials.

**Analyst:**  
  
Steve Plutschak

  
Scott Tingle